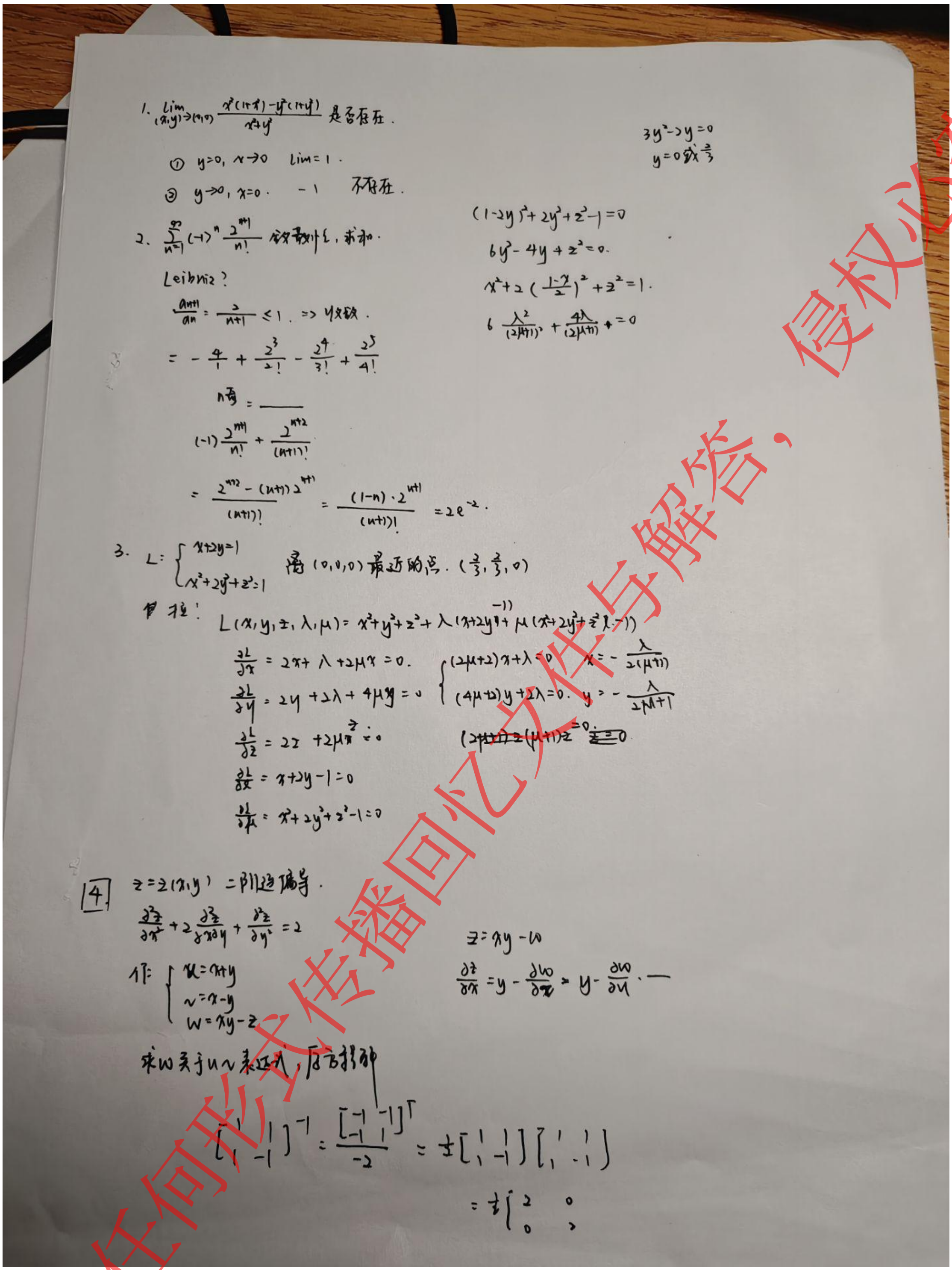


1. $\lim\limits_{(x,y)\to(0,0)} \frac{x^2(1+x)-y^2(1+y)}{x^2+y^2}$ 是否存在.

① $y=0,\ x\to 0\quad \lim=1.$

② $y\to 0,\ x=0.\quad -1$ 不存在.

$3y^2-2y=0$

$y=0$ 或 $\frac{2}{3}$

2. $\sum\limits_{n=1}^{\infty}(-1)^n\frac{2^{n+1}}{n!}$ 敛散性,求和.

Leibniz ?

$\frac{a_{n+1}}{a_n}=\frac{2}{n+1}\leq 1.\Rightarrow$ 收敛.

$=-\frac{4}{1}+\frac{2^3}{2!}-\frac{2^4}{3!}+\frac{2^5}{4!}$

n项 = ______

$(-1)\frac{2^{n+1}}{n!}+\frac{2^{n+2}}{(n+1)!}$

$=\frac{2^{n+2}-(n+1)2^{n+1}}{(n+1)!}=\frac{(1-n)\cdot 2^{n+1}}{(n+1)!}=2e^{-2}.$

$(1-2y)^2+2y^2+z^2-1=0$

$6y^2-4y+z^2=0.$

$x^2+2\left(\frac{1-x}{2}\right)^2+z^2=1.$

$6\frac{\lambda^2}{(2\mu+1)^2}+\frac{4\lambda}{(2\mu+1)}+=0$

3. $L:\begin{cases}x+2y=1\\x^2+2y^2+z^2=1\end{cases}$ 离 $(0,0,0)$ 最近的点. $(\frac{2}{3},\frac{2}{3},0)$

解: $L(x,y,z,\lambda,\mu)=x^2+y^2+z^2+\lambda(x+2y-1)+\mu(x^2+2y^2+z^2-1)$

$\frac{\partial L}{\partial x}=2x+\lambda+2\mu x=0.$

$\frac{\partial L}{\partial y}=2y+2\lambda+4\mu y=0$

$\frac{\partial L}{\partial z}=2z+2\mu z=0$

$\frac{\partial L}{\partial \lambda}=x+2y-1=0$

$\frac{\partial L}{\partial \mu}=x^2+2y^2+z^2-1=0$

$\begin{cases}(2\mu+2)x+\lambda=0 & x=-\frac{\lambda}{2(\mu+1)}\\(4\mu+2)y+2\lambda=0. & y=-\frac{\lambda}{2\mu+1}\end{cases}$

$(\cancel{2\mu+2})2(\mu+1)z=0.\ \cancel{z=0}$

$\boxed{4}$ $z=z(x,y)$ 二阶连续偏导.

$\frac{\partial^2 z}{\partial x^2}+2\frac{\partial^2 z}{\partial x\partial y}+\frac{\partial^2 z}{\partial y^2}=2$

作 $\begin{cases}u=x+y\\v=x-y\\w=xy-z\end{cases}$

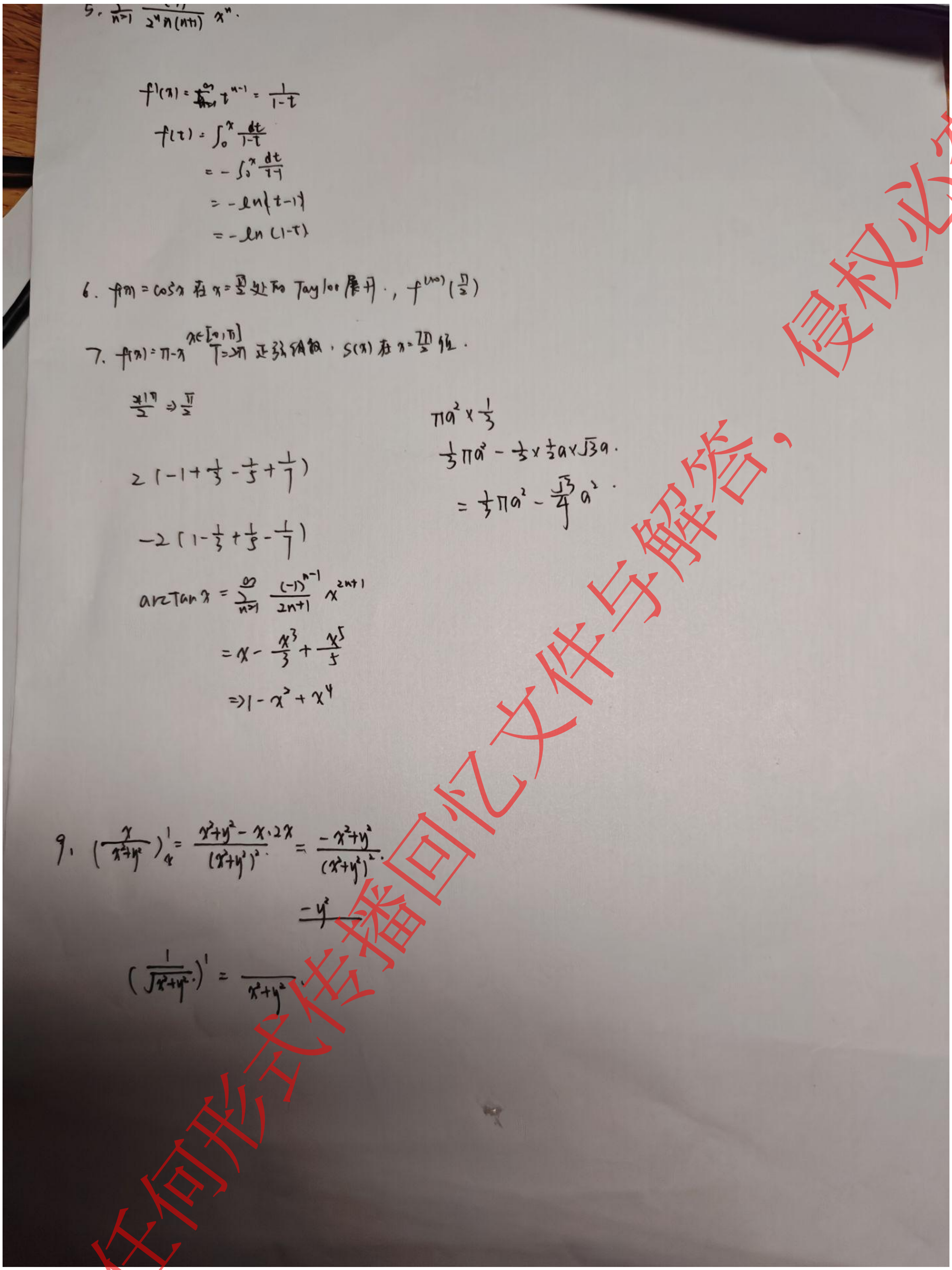
求w关于u v表达式,后面损坏

$z=xy-w$

$\frac{\partial z}{\partial x}=y-\frac{\partial w}{\partial x}=y-\frac{\partial w}{\partial u}\cdot$ —

$\begin{bmatrix}1&1\\1&-1\end{bmatrix}^{-1}=\frac{\begin{bmatrix}-1&-1\\-1&1\end{bmatrix}^T}{-2}=\frac{1}{2}\begin{bmatrix}1&1\\1&-1\end{bmatrix}\begin{bmatrix}1&1\\1&-1\end{bmatrix}$

$=\frac{1}{4}\begin{bmatrix}2&0\\0&2\end{bmatrix}$



5. $\sum_{n=1} \frac{\ }{2^n n(n+1)} x^n$.

$f'(x) = \sum_{n=1}^{\infty} t^{n-1} = \frac{1}{1-t}$

$f(t) = \int_0^x \frac{dt}{1-t}$

$= -\int_0^x \frac{dt}{t-1}$

$= -\ln|t-1|$

$= -\ln(1-t)$

6. $f(x) = \cos^2 x$ 在 $x=\frac{\pi}{2}$ 处的 Taylor 展开.，$f^{(10)}(\frac{\pi}{2})$

7. $f(x) = \pi - x$ $\overset{x\in[0,\pi]}{T=2\pi}$ 正弦级数，$S(x)$ 在 $x=\frac{7\pi}{2}$ 值.

$\frac{x+\pi}{2} \Rightarrow \frac{\pi}{2}$

$2(-1+\frac{1}{3}-\frac{1}{5}+\frac{1}{7})$

$-2(1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7})$

$\arctan x = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n+1} x^{2n+1}$

$= x - \frac{x^3}{3} + \frac{x^5}{5}$

$\Rightarrow 1 - x^2 + x^4$

$\pi a^2 \times \frac{1}{3}$

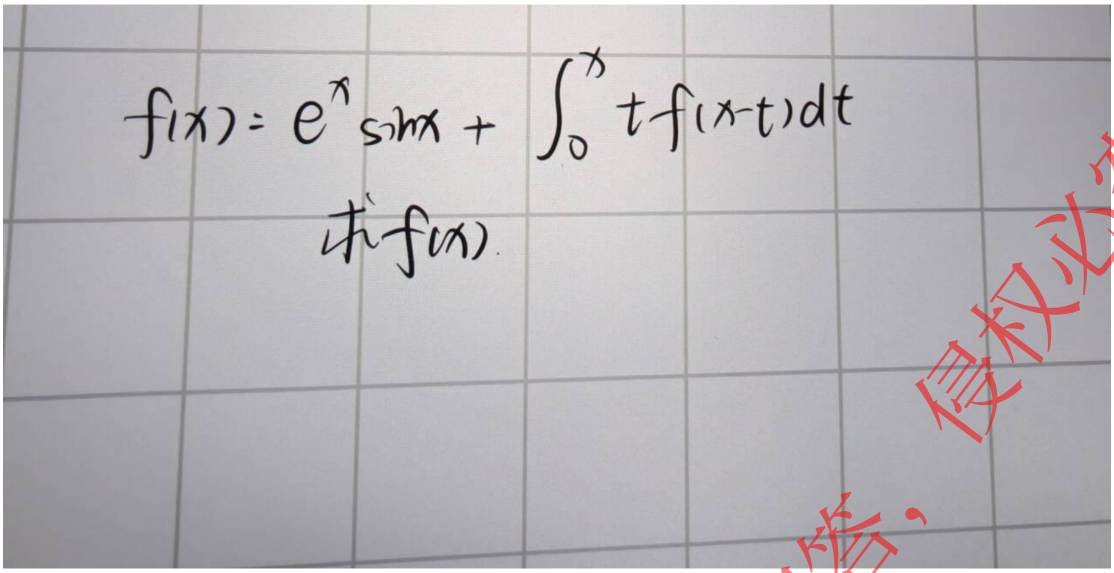
$\frac{1}{3}\pi a^2 - \frac{1}{3} \times \frac{1}{2} a \times \sqrt{3} a$.

$= \frac{1}{3}\pi a^2 - \frac{\sqrt{3}}{4} a^2$.

9. $\left(\frac{x}{x^2+y^2}\right)'_x = \frac{x^2+y^2-x\cdot 2x}{(x^2+y^2)^2} = \frac{-x^2+y^2}{(x^2+y^2)^2}$.

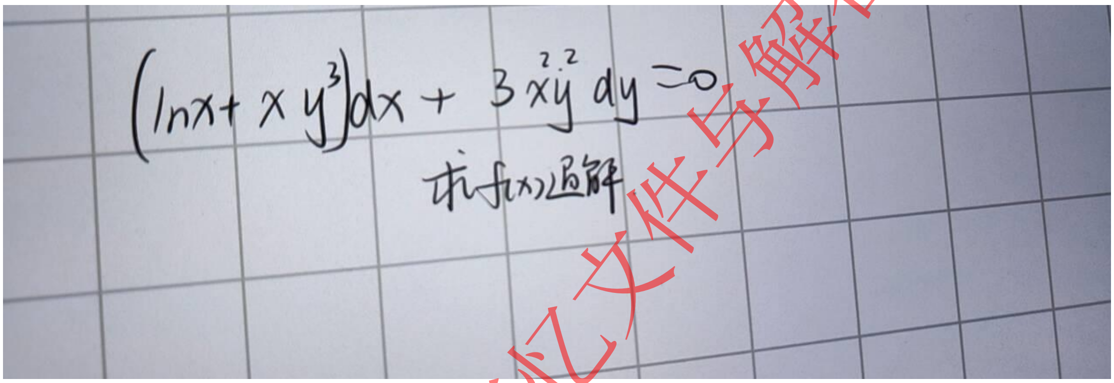
$\frac{-y^2}{\ }$

$\left(\frac{1}{\sqrt{x^2+y^2}}\right)' = \frac{\ }{x^2+y^2}$



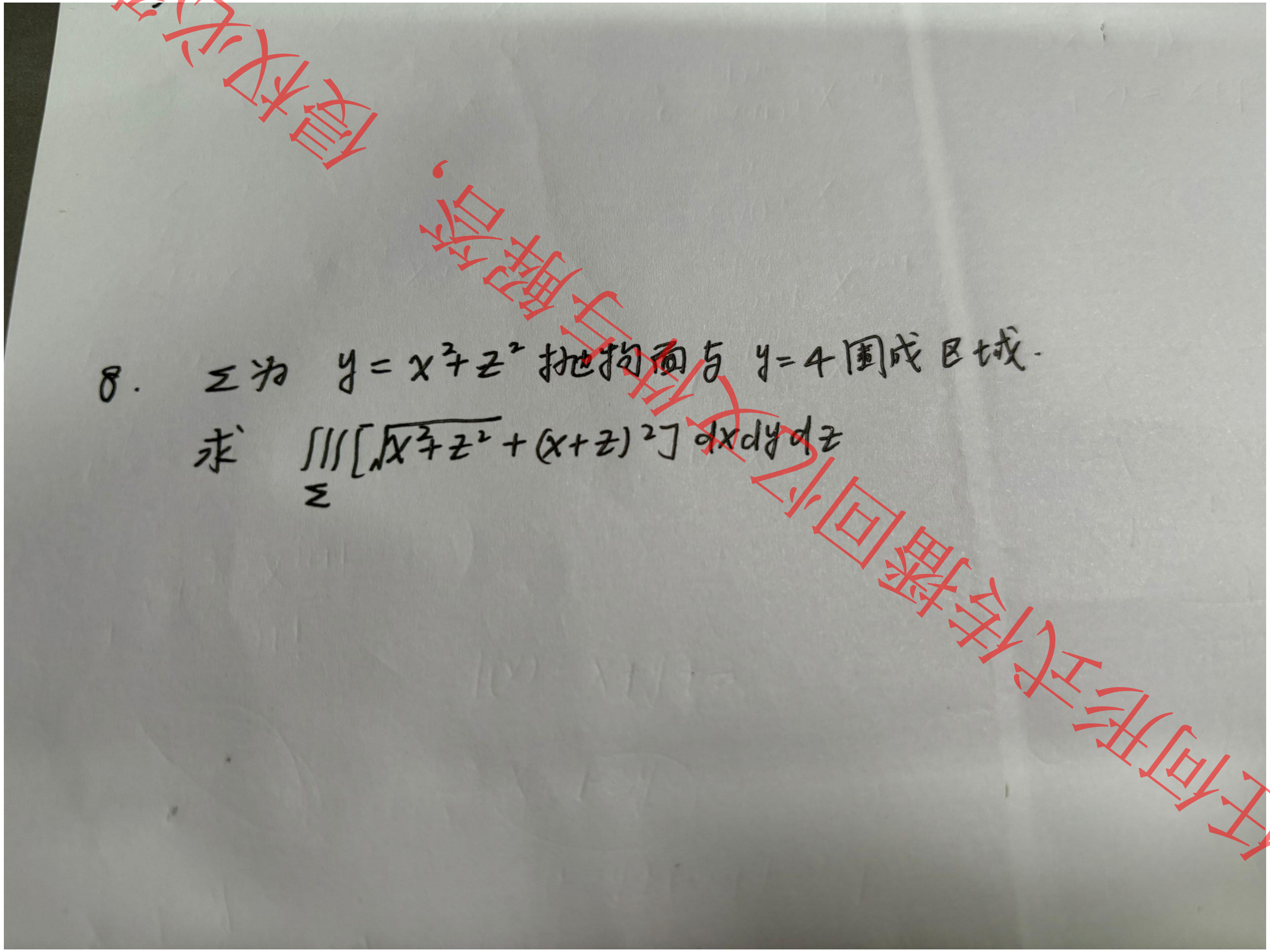
$$f(x)=e^{x}\sin x+\int_0^x t f(x-t)\,dt$$

求$f(x)$.



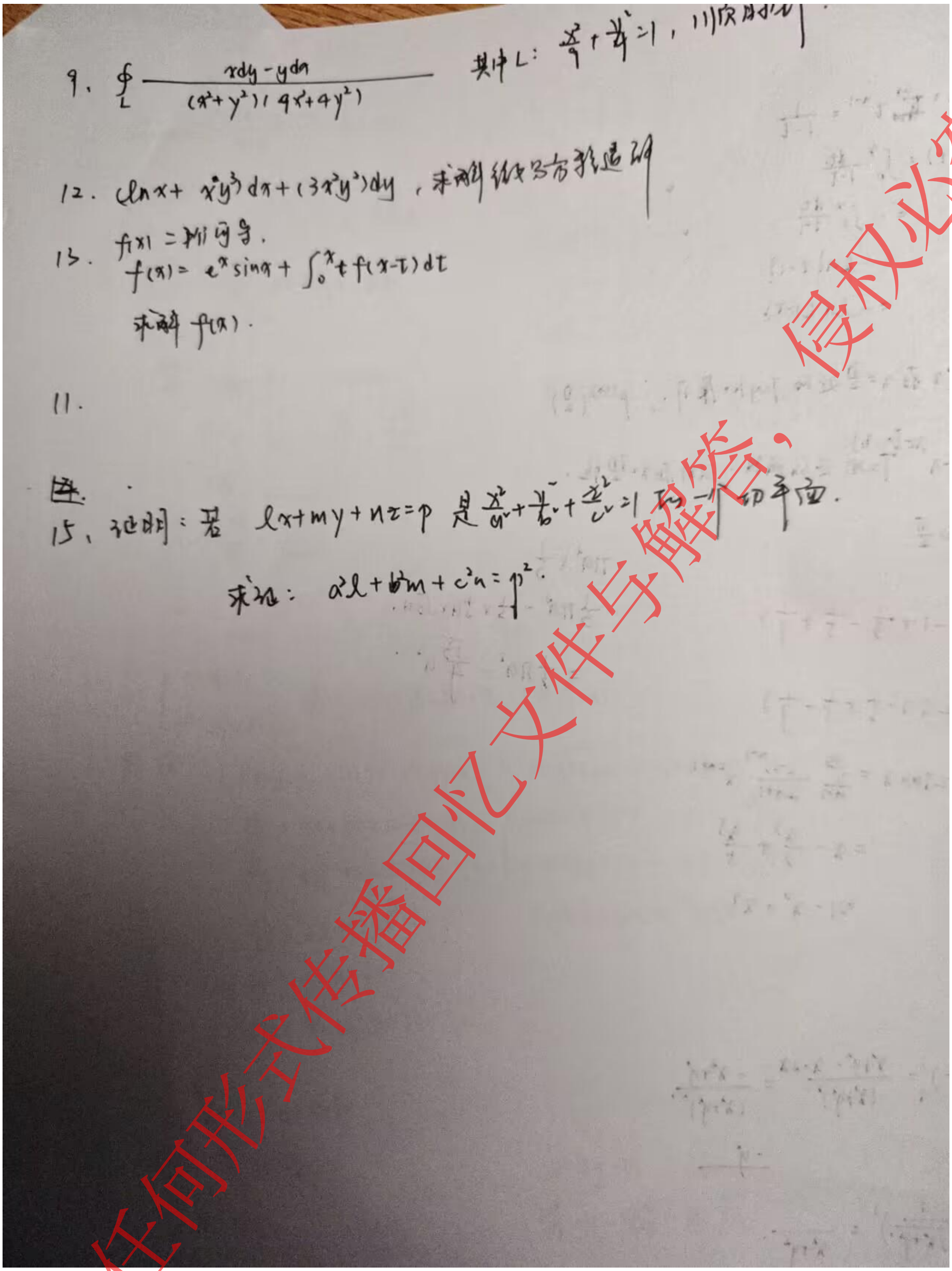
$$(\ln x+xy^3)\,dx+3x^2y^2\,dy=0$$

求$f(x)$通解



8. Σ为 $y=x^2+z^2$ 抛物面与 $y=4$ 围成区域.

求 $\iiint\limits_{\Sigma}\left[\sqrt{x^2+z^2}+(x+z)^2\right]dxdydz$



9. $\oint_L \frac{x\,dy - y\,dx}{(x^2+y^2)(4x^2+4y^2)}$ 其中 L: $\frac{x^2}{9}+\frac{y^2}{4}=1$，顺时针

12. $(\ln x + x^2y^3)dx + (3x^3y^2)dy$，求解微分方程通解

13. $f(x)$ 二阶可导.
$f(x) = e^x \sin x + \int_0^x t f(x-t)\,dt$
求解 $f(x)$.

11\.

四.
15、证明：若 $lx+my+nz=p$ 是 $\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$ 的一个切平面.
求证：$a^2l+b^2m+c^2n=p^2$.